

Machine Learning-based Late Delivery Risk Prediction in Global Supply Chain Operations

Research Paper — EDA, Predictive Modelling & Strategic Insights

Organisation: APL Logistics (KWE Group) · Programme: Unified Mentor Data Science Internship

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Dataset: 180,519 global supply chain orders · Model: Random Forest Classifier

GitHub: <https://github.com/jiteshmaurya245-dotcom/APL-Logistics-Late-Delivery-Risk>

Live App: <https://apl-logistics-late-delivery-risk-6tbthvh57qvapvpjvvtcn.streamlit.app/>

ABSTRACT

Late deliveries are a critical operational challenge in global logistics networks, causing SLA breaches, financial penalties, and customer churn. This research presents a machine learning framework to predict late delivery risk before shipment dispatch, using a dataset of 180,519 orders from APL Logistics (KWE Group). Exploratory data analysis revealed a 54.8% baseline late delivery rate with significant variation across shipping modes — First Class exhibited a 95.3% late rate versus 38.1% for Standard Class. A Random Forest Classifier achieved ROC-AUC of 0.762 and F1 score of 0.697, enabling classification of orders into Low, Medium, and High risk tiers. High-risk orders demonstrated an 84.3% actual late delivery rate, validating the model's operational utility. Key risk drivers include order profit, scheduled shipping days, and shipping pressure index. An interactive Streamlit dashboard enables real-time risk scoring and operational intervention prioritisation.

180,519

Orders analysed

0.762

ROC-AUC

84.3%

High-risk accuracy

3

Risk tiers

Project Links

GitHub Repository	https://github.com/jiteshmaurya245-dotcom/APL-Logistics-Late-Delivery-Risk
Live Deployed App	https://apl-logistics-late-delivery-risk-6tbthvh57qvapvpjvvtcn.streamlit.app/

1. Introduction

The modern global logistics sector operates under increasing pressure to deliver goods reliably, efficiently, and cost-effectively. Late deliveries represent one of the most significant operational failures, triggering SLA penalties, eroding customer trust, and inflating last-minute operational costs. Despite widespread awareness of this challenge, most logistics systems remain reactive — responding to delays only after they occur.

This research investigates a proactive approach: predicting late delivery risk before orders are dispatched. Using a dataset of 180,519 orders from APL Logistics (KWE Group), we develop, evaluate, and interpret a machine learning classification framework that assigns each order a quantitative late delivery probability and a categorical risk tier (Low, Medium, High).

1.1 Research Objectives

- Perform comprehensive EDA on 180,519 supply chain orders to identify delay patterns and risk factors.
- Engineer predictive features capturing shipping pressure, schedule tightness, and order complexity.
- Train and compare Logistic Regression, Random Forest, and Gradient Boosting classifiers.
- Assign each order a Late Delivery Probability (0–1) and categorise into Low, Medium, or High risk tiers.
- Provide explainable feature importance analysis to build operational trust in the model.
- Deploy an interactive Streamlit dashboard for real-time risk scoring and intervention planning.

1.2 Dataset Scope

The dataset comprises 180,519 order records with 40 attributes spanning payment type, product category, shipping logistics, order financials, and the binary target variable `Late_delivery_risk` (1=late, 0=on-time). Data spans five global markets: LATAM, Europe, Pacific Asia, USCA, and Africa.

2. Exploratory Data Analysis

2.1 Dataset Overview

The raw dataset contained 180,519 rows and 40 columns. Data quality was excellent — only 11 missing values (0.006%) were identified, allowing simple row-wise deletion with negligible information loss.

Attribute	Value
Total records	180,519
Total features	40
Missing values	11 (0.006%)
Target variable	Late_delivery_risk (binary 0/1)
Late deliveries	98,977 (54.8%)
On-time deliveries	81,542 (45.2%)
Class imbalance ratio	1.21:1 (mild)
Markets covered	5 (LATAM, Europe, Pacific Asia, USCA, Africa)

Table 1: Dataset overview statistics

2.2 Target Variable Distribution

The target variable exhibited a mild class imbalance with 54.8% late orders (98,977) and 45.2% on-time (81,542). This 1.21:1 ratio is manageable and does not require SMOTE. `class_weight="balanced"` was applied during model training.

2.3 Shipping Mode Analysis

Shipping mode emerged as the single most discriminative feature. Counter-intuitively, premium shipping modes exhibited the highest late rates:

Shipping Mode	Late Delivery Rate	Order Volume	Risk Level
First Class	95.3%	~45,000	Critical
Second Class	76.6%	~51,000	High
Same Day	45.7%	~23,000	Medium
Standard Class	38.1%	~62,000	Low

Table 2: Late delivery rates by shipping mode

Key finding: First Class shipping carries a 95.3% late rate — the strongest predictor in the dataset — suggesting severe systemic fulfilment constraints for this mode.

2.4 Market Region Analysis

All five global markets showed near-identical late rates (54.4%–55.2%), indicating that delay risk is not geographic but driven by order-level operational variables.

Market	Orders	Late Rate
LATAM	51,594	54.4%
Europe	50,252	55.2%

Pacific Asia	41,260	55.1%
USCA	25,799	54.8%
Africa	11,614	54.6%

Table 3: Late delivery rates by market region

2.5 Key EDA Findings

- Shipping Mode is the dominant predictor — First Class 95.3% vs Standard Class 38.1%.
- Market region is NOT predictive — all 5 markets cluster at ~54–55% late rate.
- Class imbalance is mild (1.21:1) — `class_weight="balanced"` is sufficient.
- Dataset is extremely clean — only 11 missing values in 180,519 rows.
- Financial features (profit, benefit per order) show strong correlation with delay risk.
- Tight schedules (≤ 2 days) are a significant risk amplifier.

3. Data Preprocessing & Feature Engineering

3.1 Column Removal and Leakage Prevention

Seventeen columns were removed. Two categories of removal:

Personal/identifier columns removed (no predictive value):

- Customer name, street, city, state, country
- Customer ID, Order Customer ID, Category ID, Department ID
- Latitude/Longitude (redundant with Market field), Product Name

Data leakage columns removed (critical):

- Delivery Status — directly encodes the target variable. Keeping it produces fake 100% accuracy that fails in production.
- Days for shipping (real) — only known after delivery completes. Using it makes pre-dispatch scoring impossible.

3.2 Engineered Features

Feature	Formula	Business Meaning
shipping_pressure	$\text{scheduled days} / (\text{qty} + 1)$	Time pressure per unit ordered
discount_flag	1 if discount > 0	Whether a discount was applied
profit_flag	1 if profit < 0	Whether the order is unprofitable
high_discount	1 if discount rate > 15%	High-discount order indicator
low_scheduled_days	1 if scheduled days ≤ 2	Tight schedule flag

Table 4: Engineered features

3.3 Encoding, Splitting and Scaling

- 8 categorical columns one-hot encoded → dataset expanded from 23 to 119 features.
- 80/20 stratified train/test split → 144,415 training rows, 36,104 test rows.
- StandardScaler fitted on training data only, then applied to test data.

4. Model Development & Evaluation

4.1 Model Performance Comparison

Model	ROC-AUC	Precision	Recall	F1 Score	Verdict
Logistic Regression	0.770	0.888	0.542	0.673	Baseline
Random Forest	0.762	0.739	0.659	0.697	Winner
Gradient Boosting	0.772	0.870	0.558	0.680	High precision

Table 5: Model performance comparison — test set (36,104 orders)

4.2 Model Selection: Random Forest

Random Forest was selected as the production model based on its superior F1 Score (0.697) and Recall (0.659). In logistics, Recall is critical — a missed late order (false negative) has greater operational cost than a false alarm. Random Forest's balanced precision-recall tradeoff is most operationally appropriate.

4.3 Confusion Matrix — Random Forest

	Predicted: On-time	Predicted: Late
Actual: On-time	11,704 (TN)	4,604 (FP)
Actual: Late	6,755 (FN)	13,041 (TP)

Table 6: Confusion matrix — Random Forest

5. Risk Scoring & Tier Validation

5.1 Risk Tier Classification

Risk Tier	Probability	Orders	Share	Actual Late Rate	Action
Low Risk	0.00–0.35	10,054	27.8%	33.3%	Standard processing
Medium Risk	0.35–0.65	13,280	36.8%	43.5%	Monitor & alert
High Risk	0.65–1.00	12,665	35.1%	84.3%	Immediate action

Table 7: Risk tier classification and validation results

The High Risk tier's 84.3% actual late rate is the critical validation metric — 29.5 percentage points above the 54.8% baseline. When the model flags an order as High Risk, it is correct 84 out of 100 times.

5.2 Top 10 Feature Importances

Rank	Feature	Importance	Interpretation
1	Order Profit Per Order	0.0884	Complex high-value orders face more delays
2	Benefit per order	0.0881	Financial complexity correlates with risk
3	Order Item Profit Ratio	0.0780	Margin pressure may rush fulfilment
4	Days for shipment (scheduled)	0.0582	Schedule duration affects risk directly
5	low_scheduled_days (<=2d)	0.0566	Tight schedules amplify delay probability
6	Order Item Total	0.0499	Order value linked to fulfilment complexity
7	Sales per customer	0.0498	Customer order value patterns
8	Order Item Discount	0.0492	Discounted orders may have lower priority
9	shipping_pressure index	0.0479	Days-per-unit pressure indicator
10	Order Item Discount Rate	0.0426	High discounts flag risk orders

Table 8: Top 10 feature importances — Random Forest

6. Insights & Strategic Recommendations

6.1 Key Operational Insights

Insight 1: Shipping Mode is the dominant controllable risk factor

First Class shipping carries a 95.3% late rate — the highest of any variable. This is a systemic fulfilment constraint requiring urgent operational audit.

Insight 2: Market region does not predict delay

All five markets cluster at ~54–55%, ruling out geography as a primary driver. Risk is order-specific.

Insight 3: Financial complexity correlates with delay

The top three predictors are all financial metrics. High-value complex orders likely require more coordination steps, increasing delay probability.

Insight 4: Tight schedules amplify risk

Orders with scheduled shipping ≤ 2 days have significantly higher late rates. These should be flagged for expedited processing before dispatch.

6.2 Strategic Recommendations

- Audit First Class shipping pipeline — 95.3% late rate demands immediate capacity review and process redesign.
- Deploy the predictive model for all outbound orders — score before dispatch, route High-risk to intervention queue.
- Create a tiered customer communication protocol — High-risk orders trigger proactive delay notification.
- Review tight-schedule orders (≤ 2 days) for capacity feasibility before order confirmation.
- Develop dedicated fulfilment tracks for high-value orders to reduce their disproportionate delay rate.
- Retrain the model quarterly — carrier performance and order mix evolve over time.

6.3 Business Impact

The High Risk tier identifies ~12,665 orders per 36,104-order batch with 84.3% accuracy. Proactive intervention on 50% of these correctly flagged orders could meaningfully reduce SLA penalties and improve customer satisfaction — particularly for First Class customers who are paying a premium and receiving the worst service.

7. Streamlit Dashboard

7.1 Dashboard Modules

- Risk Overview — KPI cards, risk tier donut, tier accuracy validation, probability histogram, feature importance.
- Order-Level Risk — Form-based predictor with instant probability score, gauge chart, recommended action.
- Region & Mode Analysis — Shipping mode bars, market comparison, heatmap, department risk ranking.
- Operations Action Panel — Filterable high-risk order table, CSV download, tiered action recommendations.

7.2 Technical Stack

Component	Technology
Framework	Streamlit 1.58
ML model	scikit-learn RandomForestClassifier
Data processing	pandas 2.x, NumPy 2.x
Visualisation	Plotly 6.x
Language	Python 3.10

Table 9: Technical stack

7.3 Access

The complete source code is available at the GitHub repository linked above. The dashboard is live and publicly accessible at the deployed Streamlit Community Cloud URL provided at the beginning of this document.

8. Conclusion

This research successfully demonstrated that machine learning can transform late delivery management from a reactive process into a proactive, predictive capability. Using 180,519 real-world logistics orders from APL Logistics (KWE Group), a Random Forest Classifier achieves ROC-AUC of 0.762 and F1 Score of 0.697, enabling classification of orders into Low, Medium, and High risk tiers.

The High Risk tier demonstrated 84.3% actual late delivery accuracy — providing a reliable, actionable signal for operations teams. Key delay drivers identified include order financial complexity, scheduled shipping duration, and shipping mode — with First Class shipping at 95.3% late rate being the single highest-risk factor.

The deployed Streamlit dashboard translates model outputs into an intuitive operational interface, enabling non-technical operations staff to query risk scores, filter by market and shipping mode, and download prioritised intervention lists.

8.1 Limitations and Future Work

- The model uses historical data and cannot capture real-time carrier capacity or external disruptions.
- XGBoost was not evaluated due to version constraints — future work should include this alternative.
- SHAP values would provide more granular per-order explainability than aggregate feature importance.
- Integration with a live order management system would enable true pre-dispatch scoring.
- A cost-sensitive learning approach could explicitly optimise for the asymmetric cost of false negatives.

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